Global Optimization via Softmin Energy Minimization

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Abstract

Non-convex optimization in high dimensions requires balancing gradient information with global exploration, a tension that Langevin dynamics and other gradient-informed methods address only partially. We introduce a gradient-based swarm method driven by a Softmin Energy interaction $J_\beta(\mathbf{x})$, a smooth approximation of the per-particle minimum that couples particles through their relative energies. Combining the resulting gradient flow with Brownian noise and an annealing schedule on β, we obtain a dynamics that retains the efficiency of gradient steps while inheriting the global-search behavior of swarm methods. We show, without any convexity assumption on the objective, that this interaction always partitions the ensemble into particles descending toward local minima and particles automatically repelled toward saddles and maxima, and that the same interaction caps the Freidlin-Wentzell quasipotential barrier separating adjacent basins at a level of order $1/\beta$, independently of the true barrier height, which is exponentially cheaper than the barrier faced by an overdamped Langevin particle in the same landscape. Numerical experiments on double-well and high-dimensional Ackley benchmarks confirm the analysis and show improved convergence over Simulated Annealing, Gradient-Enhanced Particle Swarm Optimization, and Gradient-Aware Consensus Based Optimization.

1 Introduction

Non-convex optimization problems are notoriously difficult, as the presence of multiple local minima can trap gradient-based algorithms. Existing approaches such as gradient descent perform well for convex functions with a single minimum but will likely converge to suboptimal local minima in more complex landscapes. This challenge is especially prominent in machine learning, where training deep neural networks on non-convex losses often leads to inconsistent performance across random initializations. Similarly, in reinforcement learning, escaping sharp local minima remains a key bottleneck for achieving state-of-the-art results.

A classical strategy to address this issue is to add a noise term to the dynamics, allowing the system to escape local minima and leading to stochastic gradient–based methods, for example Langevin dynamics [10]. Stochastic gradient-based methods for optimization, while effective, typically exhibit poor transition times between local minima. The escape rate of a particle subject to Gaussian noise

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from a local minimum depends exponentially on the height of the potential barrier surrounding that minimum [5]. Methods that improve convergence rates over gradient-based algorithms often rely on additional assumptions about the objective function f. For example, if f admits a decomposition as a linear combination of simpler functions ψ_i, i.e., $f(x) = \sum_i \psi_i(x)$, then the Method of Moments [22] can be employed to converge to the global minimum. This type of decomposition is particularly useful in areas such as signal processing and machine learning, where complex functions are represented as sums of basis functions, such as Fourier series or wavelets [29]. In more general settings where such structure is not available, Simulated Annealing and its variants [7, 35] are widely used, leveraging a decaying temperature schedule to shift the dynamics from noisy exploration to refined local search.

A natural extension of stochastic gradient-based approaches is the use of multiple particles to explore the search space in parallel. For instance, when particles evolve independently under Langevin dynamics, the expected transition times between local minima decrease compared to single-particle dynamics purely due to statistical averaging over multiple independent trajectories. Moreover, allowing particles to cooperate rather than evolve independently can further improve exploration and convergence. The idea of particle cooperation is often central in Metaheuristic methods [4, 36]. Such methods provide a broad framework for global optimization and have demonstrated strong empirical performance, although establishing rigorous theoretical convergence guarantees is often challenging. Many such methods employ a particle-swarm-like paradigm [19, 27], in which multiple interacting particles explore the search space in parallel to locate global minimizers.

The global optimization method proposed in this paper builds on these ideas by combining stochastic gradient techniques with cooperative multi-particle dynamics. We employ the well-known Softmin Energy functional J_β, which smoothly interpolates between the average and the hard minimum of the function values across particle positions. This functional acts as the source of interaction among particles, whose evolution is governed by its gradient, driving the system toward more effective exploration and optimization.

The Softmin function provides a natural way to interpolate between uniform weighting and hard minima via temperature $\beta > 0$. A common use of the Softmin function is to define the Gibbs distribution, where the probability of sampling x_i is given by:

$$\mathbb{P}[i] = \frac{\exp(-\beta x_i)}{\sum_{j=1}^n \exp(-\beta x_j)},$$

with $\beta \to \infty$ concentrating on $\min_i x_i$ and $\beta \to 0$ uniform [6]. This underlies stochastic optimization [1, 31, 16, 15] and exploration in Reinforcement Learning [24, 18].

Our proposed method is a stochastic gradient-based particle swarm approach that leverages a Softmin-type energy functional to effectively incorporate gradient information from the particles. Consider a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. The objective is to approximately solve the global minimization problem $\min_{x\in\mathbb{R}^d} f(x)$. Given a population of n particles $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \in \mathbb{R}^d$ for all $i \in [n]$, the Softmin Energy is defined as:

$$J_\beta(\mathbf{x}) = \sum_{i=1}^n \frac{\exp(-\beta f(x_i))}{\sum_{j=1}^n \exp(-\beta f(x_j))} f(x_i), \tag{1}$$

which provides a smooth approximation of the empirical minimum among the $f(x_i)$. The parameter $\beta > 0$ controls the approximation's sharpness: larger values yield a closer approximation to the hard minimum $\min_i f(x_i)$, while $\beta \to 0$ recovers the average $\frac{1}{n}\sum_{i=1}^n f(x_i)$.

To minimize J_β we consider the stochastic gradient flow in the particle space $\mathbb{R}^{nd}$:

$$\mathrm{d}\mathbf{x}_t = -n\nabla J_{\beta_t}(\mathbf{x}_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t, \tag{2}$$

where $(B_t)_{t\geq 0}$ represents a standard nd-dimensional Brownian motion and $\sigma > 0$ introduces stochasticity to promote exploration and escape from local minima.

1.1 Contribution

The main novelty of this work is the rigorous analysis of stochastic gradient-based dynamics driven by the Softmin Energy functional, $J_\beta(\mathbf{x})$. In this paper, we prove that the resulting inter-particle interaction mechanism provides strict theoretical advantages over standard gradient-based and non-interacting methods for non-convex global optimization.

Dynamic Ensemble Partitioning and Accelerated Convergence The interaction induced by ∇J_β naturally partitions the particle ensemble via an effective learning rate, $A_{t,k}$. At every stable equilibrium, and without any convexity assumption on f, particles below the Softmin threshold ($A_{t,k} > 0$) sit at local minima, while trapped, high-energy particles ($A_{t,k} < 0$) are automatically repelled to hill-climb and explore, with at least one particle always at a local minimum (Theorem 2.1). Along the trajectory, the sign of $A_{t,k}$ is not conserved for any single particle; instead, we prove that the running best objective value across the whole ensemble decreases at least as fast as ordinary gradient descent whenever f satisfies a Polyak–Łojasiewicz condition, for any – possibly time-varying – inverse-temperature schedule (Theorem 2.4).

Provable Topological Barrier Flattening We demonstrate that the Softmin dynamics do not merely reduce effective potential barriers, but analytically cap them. By integrating the effective drift along an explicit escape channel, we prove that arbitrarily high physical barriers separating adjacent basins are capped, at the level of the Freidlin-Wentzell quasipotential, to a constant of order $1/\beta$, uniformly in the true barrier height (Theorem 3.2). This bound holds unconditionally on the quasipotential itself and is exponentially cheaper than the exact small-noise exit exponent of an overdamped Langevin particle in the same landscape; turning it into a guarantee on the expected transition time additionally requires the metastable-domain hypotheses made precise in Corollary 3.3. This structural landscape modulation is a direct consequence of the continuous Softmin repulsion, distinguishing our approach from generic stochastic swarm methods and Simulated Annealing.

1.2 Related work

Metaheuristic optimization methods Metaheuristic optimization methods include Consensus-Based Optimization (CBO) [8, 13, 25, 9], Genetic Algorithms [17], Ant Colony Optimization [12], and Particle Swarm Optimization (PSO) [20, 23]. Traditional particle-swarm approaches typically operate without explicit gradient information, relying instead on position-based updates to coordinate particles toward promising regions. When reliable gradient information is available, leveraging it enables convergence rates comparable to standard stochastic gradient descent, which gradient-agnostic metaheuristics cannot achieve.

Recent gradient-based swarm methods [28, 7, 32, 9, 23], have made progress by incorporating gradient information into multi-particle dynamics. However, our approach differs fundamentally in its mechanism for inducing exploration and escaping barriers. Rather than consensus-based averaging or general gradient-weighted potentials, we employ the Softmin energy functional to explicitly create a minimizing/maximizing partition of the particle ensemble: particles below the Softmin threshold are drawn toward local minima via gradient descent, while particles above the threshold are repelled toward saddles and maxima. This automatic role assignment (Theorem 2.1) simultaneously reduces effective potential barriers (Theorem 3.2) and forces suboptimal particles into exploratory regimes without requiring manual basin-switching heuristics.

To the best of our knowledge, no other metaheuristic employs the Softmin energy functional, computed as the softmin of function values at particle positions, as the sole source of inter-particle interaction. This design enables provable faster escape times from local maxima and provable linear convergence rates to minima in the small-noise limit, as demonstrated in Section 3 and validated experimentally on benchmark landscapes where barriers dominate the hardness.

Langevin-based sampling and interacting-particle methods Stochastic Gradient Langevin Dynamics (SGLD) [34] and its non-asymptotic analysis for non-convex optimization [26] study a single diffusion whose drift is the gradient of the objective itself, with guarantees governed by the spectral gap of the associated Gibbs measure; when multiple particles are used, interaction typically enters only through averaging over independent runs. Interacting Particle Langevin Dynamics (IPLD) [2, 11] instead couples an ensemble of particles through their empirical measure, using this mean-field interaction to build unbiased or stabilized estimators of the marginal likelihood gradient in latent-variable models. Our construction is closer in spirit to IPLD, in that it also relies on genuine ensemble-level coupling rather than independent copies, but the interaction itself is different in kind: instead of pooling particles through an empirical average, the Softmin energy J_β couples particles through their relative energies, producing a state-dependent, sign-changing effective learning rate $A_{t,k}$ that deterministically assigns each particle a minimizing or maximizing role (Theorem 2.1). It is this competitive, temperature-sharpened coupling, rather than mean-field averaging, that produces

the barrier-capping mechanism of Theorem 3.2, and it is aimed directly at global optimization of f rather than at marginal-likelihood estimation.

Connection to Statistical Physics and Functional Design Our approach shares conceptual roots with population-based statistical physics methods such as Population Annealing (PA) [33], which also employs a Gibbs-weighted ensemble to locate global ground states. However, while PA relies on discrete demographic selection, duplicating low-energy particles and eliminating high-energy ones prior to local Markov Chain Monte Carlo steps, our method maintains a fixed population size and drives exploration through continuous dynamical repulsion. This active exploration mechanism differentiates our Softmin Energy functional from the standard log-sum-exp (or free-energy) functional, $F_\beta(\mathbf{x}) = -\beta^{-1} \log \sum e^{-\beta f(x_i)}$. The gradient of the log-sum-exp formulation, $\nabla_{x_k} F_\beta(\mathbf{x}) = w_k \nabla f(x_k)$, strictly preserves gradient descent, causing sub-optimal particles with small weights to simply stagnate. In contrast, the gradient of our proposed Softmin functional, $J_\beta(\mathbf{x}) = \sum w_i f(x_i)$, natively incorporates the sign-changing effective learning rate, $A_{t,k}$. Consequently, rather than discarding trapped particles, the geometry of the Softmin functional actively destabilizes their local minima, forcing high-energy particles to hill-climb and escape via the topological barrier-flattening mechanism formalized in our theoretical analysis.

Softmin Energy Theoretical work on the Gibbs Softmin (or Boltzmann Softmin) has primarily focused on its properties as a smooth approximation to the minimum operator [6], convergence of softmax policy gradients in reinforcement learning [21], and its role in exploration–exploitation trade-offs via temperature scheduling [30]. To our knowledge, this is the first work employing J_β as an optimization objective for non-convex global search. The effective barrier reduction via particle cooperation, analyzed in Theorem 3.2, is previously unstudied.

1.3 Structure of the paper

In Section 2, we analyze the deterministic gradient flow dynamics obtained by setting $\sigma = 0$ in (2) in detail. Section 2.1 characterizes the stable equilibria of this flow with no convexity assumption on f: every particle sits at a critical point of f, the sign of its effective learning rate fixes whether that critical point is a minimum or a maximum, and at least one particle always resides at a local minimum. Section 2.2 then shows that, whenever f satisfies a Polyak–Łojasiewicz condition, the best objective value across the ensemble decreases at least as fast as ordinary gradient descent, for any inverse-temperature schedule.

In Section 3, we examine the noisy dynamics (2) at fixed inverse temperature, showing that the Softmin interaction caps the Freidlin-Wentzell quasipotential barrier separating adjacent basins at a level of order $1/\beta$, independently of the true barrier height – exponentially cheaper than the exact small-noise exit exponent of an overdamped Langevin particle in the same landscape.

In Section 4, we conduct numerical experiments to validate the method's effectiveness in escaping local minima and identifying global optima. Comparisons with Simulated Annealing, Gradient-Aware CBO [9], and Gradient-Enhanced PSO [23] show better performance, particularly when potential barriers exceed a critical threshold.

The proofs of lemmas and theorems throughout the paper are deferred to Appendix B. Details on experimental settings can be found in Appendix C.

2 Gradient Flow dynamics for Softmin Energy

This section focuses on the deterministic gradient flow dynamics

$$\mathrm{d}\mathbf{x}_t = -n \nabla J_\beta(\mathbf{x}_t) \mathrm{d}t, \tag{3}$$

where $\mathbf{x}_t = (x_1(t), \ldots, x_n(t))$ denotes the state of the particle system and the Softmin energy $J_\beta(\mathbf{x})$ is defined in (1). The scaling factor n, which can be absorbed via a time change, ensures that the subsequent derivatives and governing equations remain well-posed in the mean-field limit $n \to \infty$. This continuous-time system performs gradient descent on the Softmin landscape, which is formulated such that its minimization inherently drives the ensemble toward the minima of the objective function f.

Subsequent analysis characterizes the stable equilibria of this flow without any convexity assumption on f: at every such equilibrium, each particle sits at a critical point of f, the sign of its effective learning rate $A_{t,k}$ fixes whether that critical point is a minimum or a maximum, and at least one particle resides at a local minimum. We further show that the best objective value across the ensemble decreases at least as fast as ordinary gradient descent under a Polyak–Łojasiewicz condition, for any (possibly time-varying) inverse temperature schedule.

To formally analyze these dynamics, it is necessary to first evaluate the gradient of nJ_β with respect to the individual particle coordinates. Let $w_k = e^{-\beta f(x_k)}(\sum_j e^{-\beta f(x_j)})^{-1}$ denote the Gibbs weight of particle x_k, and define the dynamic pre-factor $A_{t,k} = nw_k\,[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]$. The gradient of $nJ_\beta(\mathbf{x})$ with respect to particle x_k is computed in Lemma B.1 and is given by:

$$\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_{t,k}\nabla f(x_k), \tag{4}$$

so that (3) reads componentwise as $\dot{x}_k = -A_{t,k}\nabla f(x_k)$. The exact second-order partial derivatives and corresponding Hessian blocks, which are required for the subsequent local stability analysis, are deferred to Appendix B.1.

This formulation reveals a critical property of the Softmin gradient. The dynamic pre-factor $A_{t,k} = nw_k\beta\,[1/\beta - f(x_k) + J_\beta(\mathbf{x})]$ dictates the effective direction of the gradient flow for each particle x_k, acting as a state-dependent learning rate. According to the dynamics in (3), if a particle evaluates to a landscape value $f(x_k) < J_\beta(\mathbf{x}) + 1/\beta$, the pre-factor $A_{t,k}$ is strictly positive, and the particle performs standard gradient descent on f; the particle is then said to be *minimizing*. Conversely, if $f(x_k) > J_\beta(\mathbf{x}) + 1/\beta$, the pre-factor becomes negative, effectively reversing the flow, and the particle is *maximizing*, executing gradient ascent on f. This mechanism theoretically guarantees that particles trapped in sub-optimal local minima are actively repelled and forced to explore the landscape, whereas particles at comparatively lower energy states continue to minimize f.

Two elementary facts drive the equilibrium analysis that follows. First, summing the pre-factor over all particles and using $\sum_k w_k = 1$ and $\sum_k w_k f(x_k) = J_\beta(\mathbf{x})$,

$$\sum_{k=1}^{n} A_{t,k} = n \qquad \text{for every } \mathbf{x} \in \mathbb{R}^{nd}, \tag{5}$$

so at least one coefficient is always strictly positive. Second, at any configuration where every particle sits at a critical point of f, the mixed second partial derivatives of J_β vanish identically, since each mixed Hessian block is an outer product carrying a factor $\nabla f(x_k)$ (Lemma B.1); the Hessian of nJ_β is then block diagonal.

2.1 Stationary points of the dynamics

This subsection characterizes the stable equilibria of the gradient flow (3), with *no convexity assumption whatsoever* on the objective function f. Theorem 2.1 shows that at any such equilibrium every particle sits at a critical point of f with a strictly nonzero pre-factor, the sign of that pre-factor fixes whether the critical point is a minimum or a maximum, and at least one particle always resides at a local minimum. Consequently, the Softmin formulation guarantees both the exploitation of deep minima and the systematic destabilization of higher-energy critical points, without requiring any local geometric hypothesis on f.

Theorem 2.1. *Let $f \in C^2(\mathbb{R}^d)$, let $\beta > 0$, and let $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_n)$ be a locally asymptotically stable equilibrium of* (3)*. Then, for every $k \in \{1, \ldots, n\}$, evaluated at $\bar{\mathbf{x}}$:*

1. $\nabla f(\bar{x}_k) = 0$*;*
2. $A_{t,k} \neq 0$*;*
3. *the sign of $A_{t,k}$ fixes the type of $\bar{x}_k$:*

   $A_{t,k} > 0 \implies \bar{x}_k$ *is a local minimum of* f, $\qquad A_{t,k} < 0 \implies \bar{x}_k$ *is a local maximum of* f,

   with $A_{t,k} > 0 \iff f(\bar{x}_k) < J_\beta(\bar{\mathbf{x}}) + 1/\beta$.

Moreover, by (5)*, at least one particle sits at a local minimum of f.*

Proof. The rigorous derivation of these stability conditions is deferred to Appendix B.1. □

Corollary 2.2 (Sufficient condition for stability)**.** *Let* $\bar{\mathbf{x}} = (\bar{x}_1, \dots, \bar{x}_n)$ *satisfy* $\nabla f(\bar{x}_k) = 0$ *for every* k*, and assume that for each* k *either*

$$A_{t,k} > 0 \text{ and } \nabla^2 f(\bar{x}_k) \succ 0, \qquad \text{or} \qquad A_{t,k} < 0 \text{ and } \nabla^2 f(\bar{x}_k) \prec 0,$$

evaluated at $\bar{\mathbf{x}}$*. Then* $\nabla^2(nJ_\beta)(\bar{\mathbf{x}}) \succ 0$*; in particular* $\bar{\mathbf{x}}$ *is a strict local minimum of* nJ_β *and a locally asymptotically stable equilibrium of* (3)*.*

Proof. By Lemma B.1 the Hessian at a fully critical configuration is block diagonal with blocks $A_{t,k}\nabla^2 f(\bar{x}_k)$, and each hypothesis makes the corresponding block positive definite. □

The deterministic flow can therefore become trapped at any equilibrium consisting of particles at local minima and local maxima of f, not only at full consensus on a single (possibly suboptimal) minimum; the noisy dynamics analyzed next provide a mechanism for the ensemble to escape such traps.

2.2 Descent of the ensemble minimum

The instantaneous sign of the pre-factor $A_{t,k}$ is not conserved for a fixed particle, since it depends on every coordinate through J_β. The invariant object is instead the running best value across the ensemble,

$$m(t) := \min_{1 \le k \le n} f(x_k(t)),$$

which we now control for a general, possibly time-dependent, inverse-temperature schedule $\beta_t > 0$. Write $A_{t,k}$ for the pre-factor evaluated at inverse temperature β_t and configuration $\mathbf{x}(t)$.

Definition 2.3 (Polyak–Łojasiewicz condition)**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ satisfies the *Polyak–Łojasiewicz (PL) condition* on a set $S \subseteq \mathbb{R}^d$ with constant $\mu > 0$ if

$$\tfrac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f_\star\big), \qquad f_\star := \inf f,$$

for every $x \in S$.

Theorem 2.4 (Descent of the ensemble minimum)**.** *Let* $f \in C^1(\mathbb{R}^d)$ *have locally Lipschitz gradient, let* $\beta_t > 0$ *be measurable, and let* $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))$ *be a global solution of*

$$\dot{x}_k(t) = -A_{t,k}\,\nabla f(x_k(t)), \qquad k = 1, \dots, n.$$

Suppose f *satisfies the Polyak–Łojasiewicz condition (Definition 2.3) with constant* $\mu > 0$ *on the sublevel set* $S_0 := \{x : f(x) \le m(0)\}$*. Then* m *is nonincreasing and*

$$m(t) - f_\star \le e^{-2\mu t}\big(m(0) - f_\star\big), \qquad t \ge 0.$$

Proof. The proof is deferred to Appendix B.1. □

The estimate compares the best value in the ensemble against ordinary gradient flow; it does not assign a permanent role to any designated particle, and it holds for any positive schedule β_t, in every dimension.

Remark 2.5 (Global existence). Theorem 2.4 presupposes a global solution. Since m is nonincreasing, every particle attaining the running minimum stays in the compact sublevel set S_0 when f is coercive; the theorem controls these particles only. Maximizing particles ($A_{t,k} < 0$) are damped by an exponentially small Gibbs weight, but their trajectories are not controlled by this estimate, and global existence must be assumed or established separately.

3 Hitting Times

This section analyzes the stochastic gradient flow dynamics, given by:

$$\mathrm{d}\mathbf{x}_t = -n\nabla J_\beta(\mathbf{x}_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t, \tag{6}$$

where the term $\sqrt{2\sigma}\mathrm{d}B_t$ introduces isotropic nd-dimensional Brownian noise. Componentwise,

$$\mathrm{d}x_k(t) = -A_{t,k}\nabla f(x_k(t))\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t^k, \tag{7}$$

so $A_{t,k}$ acts as a state-dependent, sign-changing effective learning rate for particle x_k, as established in Section 2. The rigorous exit analysis below is carried out for *fixed* β and physical dimension $d = 1$; the time-inhomogeneous (annealed) case $\beta = \beta_t$ requires a separate treatment and is used only heuristically in Section 4. The ensemble-descent estimate of Section 2.2, by contrast, holds for any positive schedule and in every dimension.

3.1 One-dimensional basin exit and post-saddle passage

Fix $n \geq 2$, $\beta > 0$, and $d = 1$. Consider three consecutive nondegenerate critical points of f,

$$z_- < s < z_+,$$

with $z_\pm$ local minima and s the separating local maximum, and assume

$$f'(x) > 0 \text{ on } (z_-, s), \qquad f'(x) < 0 \text{ on } (s, z_+). \tag{8}$$

Set the relative energy and the barrier height

$$r(x) := f(x) - f(z_-), \qquad h := r(s) > 0, \qquad \rho_+ := r(z_+) < h. \tag{9}$$

No sign is imposed on ρ_+; the case of interest is $\rho_+ < 0$, in which z_+ is a strictly deeper minimum than z_-.

We rely on the Freidlin-Wentzell large deviation theory for exit times of small-noise diffusions from metastable domains, summarized in Appendix A: under the standard hypotheses of that theory, the exit time τ_D^σ from a metastable domain D satisfies $\lim_{\sigma\downarrow 0}\sigma\log\mathbb{E}_{\mathbf{x}}\tau_D^\sigma = V_\beta(\mathbf{x}, \partial D)$, where V_β is the Freidlin-Wentzell quasipotential associated with (6).

To build an admissible escape path, freeze $n-1$ particles at z_- and let one particle move:

$$\Gamma(x) := (x, z_-, \ldots, z_-). \tag{10}$$

Factoring $e^{-\beta f(z_-)}$ out of the Gibbs weights gives $J_\beta(\Gamma(x)) = f(z_-) + u(r(x))/n$, that is, the exact identity

$$nJ_\beta(\Gamma(x)) = nf(z_-) + u(r(x)), \qquad u(r) := \frac{n\, r\, e^{-\beta r}}{n - 1 + e^{-\beta r}}, \quad r \in \mathbb{R}, \tag{11}$$

where u is the *reduced potential*. The additive constant $nf(z_-)$ does not enter $\nabla(nJ_\beta)$, and both the action and the quasipotential depend on nJ_β only through its increments. Note that u is defined on all of $\mathbb{R}$, since $r(x)$ is negative wherever f drops below $f(z_-)$.

The reduced potential also records the sign of the drift of the active particle, and this is what fixes the geometry of the channel.

Lemma 3.1 (Channel drift and shape of the reduced potential)**.** *Let $A_{t,1}$ denote the pre-factor of the active particle along the channel. Then, for every $x \in \mathbb{R}$,*

$$A_{t,1}(\Gamma(x)) = u'(r(x)) = \frac{n\, e^{-\beta r(x)}\, q(r(x))}{\left(n - 1 + e^{-\beta r(x)}\right)^2}, \qquad q(r) := (n-1)(1-\beta r) + e^{-\beta r}. \tag{12}$$

The function q is strictly decreasing on $\mathbb{R}$ and has a unique zero r_c, which satisfies

$$\frac{1}{\beta} < r_c < \frac{n}{(n-1)\beta}. \tag{13}$$

Consequently u is strictly increasing on $(-\infty, r_c]$ and strictly decreasing on $[r_c, \infty)$, and r_c is exactly the relative-energy level at which the active particle switches from minimizing to maximizing, i.e. the level at which $f(x) - J_\beta(\Gamma(x)) = 1/\beta$.

Proof. See Appendix B.2. □

We abbreviate the two relevant channel heights by

$$u_c := u(r_c) = \max_{r\in\mathbb{R}} u(r), \qquad u_h := \max_{0\le r\le h} u(r) = u\big(\min\{h, r_c\}\big). \tag{14}$$

Theorem 3.2 (One-dimensional basin exit and post-saddle passage)**.** *Under the assumptions* (8)–(9) *the following hold.*

1. ***(Uniform barrier cap.)*** *The channel barrier is bounded independently of the physical barrier h and of f:*
$$u_c = \max_{r\in\mathbb{R}} u(r) < \frac{n}{e(n-1)\beta}. \tag{15}$$
In particular both the barrier u_c and, by (13)*, the switching level r_c are of order β^{-1}, uniformly in h.*

2. ***(Initial basin exit: quasipotential bound.)*** *The minimum action among channel paths joining $\Gamma(z_-)$ to $\Gamma(s)$ equals u_h. Consequently*
$$V_\beta\big(\Gamma(z_-), \Gamma(s)\big) \le u_h < \min\{h, r_c\} \le h. \tag{16}$$

3. ***(Post-saddle well and passage.)*** *Assume*
$$r_c < h. \tag{17}$$
Then $\Gamma(s)$ is a strict local minimum of nJ_β, with channel barrier $u_c - u(h)$ on the z_- side (realized at x_m). The behavior on the z_+ side depends on where ρ_+ falls relative to r_c:

   - *If $\rho_+ < r_c$, there is a unique $\xi_+ \in (s, z_+)$ with $r(\xi_+) = r_c$; the channel well at $\Gamma(s)$ is symmetric, with barrier $u_c - u(h)$ on this side as well, the minimum action among channel paths joining $\Gamma(s)$ to $\Gamma(\xi_+)$ equals $u_c - u(h)$, and therefore*
   $$V_\beta\big(\Gamma(s), \Gamma(\xi_+)\big) \le u_c - u(h) < u_c; \tag{18}$$
   beyond ξ_+ the deterministic channel dynamics drive the active particle to z_+.
   - *If instead $\rho_+ \ge r_c$, then $r(x) \in [r_c, h)$ throughout (s, z_+); since u is strictly decreasing on $[r_c, \infty)$ and r is strictly decreasing on (s, z_+), the channel potential $u(r(x))$ increases monotonically from $u(h)$ at s to $u(\rho_+)$ at z_+, with no interior peak: on this side $\Gamma(s)$ is not separated from $\Gamma(z_+)$ by any channel barrier, and no point ξ_+ exists.*

   If instead $h \le r_c$, then $\Gamma(s)$ is not a local minimum of nJ_β and no post-saddle well forms.

Proof. See Appendix B.2. □

Corollary 3.3 (Exit-time consequence, conditional)**.** *Suppose in addition that z_- admits a metastable domain $D_- \subset \mathbb{R}^n$ for the flow* (3)*, in the sense that $\Gamma(z_-)$ is the unique attractor of D_- for the noiseless flow and the hypotheses of the Freidlin-Wentzell exit theorem (Appendix A) hold on D_-, with $\Gamma(x_m) \in \partial D_-$ realizing the minimal quasipotential to ∂D_-, where $x_m \in (z_-, s]$ is the unique point with $r(x_m) = \min\{h, r_c\}$. Then*

$$\lim_{\sigma\downarrow 0} \sigma \log \mathbb{E}_{\Gamma(z_-)} \tau^\sigma_{D_-} = V_\beta\big(\Gamma(z_-), \partial D_-\big) \le u_h < h.$$

We do not construct D_- or verify these hypotheses here; Theorem 3.2 supplies only the quasipotential bound (16)*, which holds unconditionally.*

Remark 3.4 (Low-barrier regime: acceleration without sign reversal). By (12), $A_{t,1}(\Gamma(x)) > 0$ precisely when $r(x) < r_c$. Hence $u_h = u(h)$, i.e. $\min\{h, r_c\} = h$, exactly in the regime $h \le r_c$, and there the drift of the active particle is minimizing along the entire crossing: the sign-reversal mechanism is never activated, the crossing is made entirely against the drift, and by Theorem 3.2(3) no Softmin-induced well appears at s. The acceleration is nonetheless strict, since $u(r)/r = ne^{-\beta r}/(n-1+e^{-\beta r}) < 1$ for $r > 0$ gives $u(h) < h$. In this regime the gain comes purely from the Gibbs-weight damping of the leading particle's contribution to nJ_β, not from a sign reversal. By (13) the regime is $h \lesssim \beta^{-1}$; the sign-reversal mechanism is what handles barriers beyond that scale.

Remark 3.5 (Interpretation and scope). Part 1 shows that, along this channel, an arbitrarily high physical barrier h is flattened to a level of order β^{-1}: as h grows the active particle receives exponentially small Gibbs weight, so the barrier saturates rather than growing with h. Part 2 gives a strict full-system *upper* bound on the exit exponent through an explicit admissible path; because V_β is an infimum over all paths in $\mathbb{R}^n$, the channel construction can only over-estimate it. It is a bound on the Freidlin-Wentzell quasipotential itself, established by exhibiting an admissible path; it does not yet assert anything about $\mathbb{E}\tau^\sigma$, for which see Corollary 3.3. Part 3 addresses the post-saddle obstruction: a high barrier turns the physical maximum s into a Softmin-induced well. When $\rho_+ < r_c$ this well is symmetric, with depth $u_c - u(h)$ strictly below the initial barrier u_c, so it is not a worse obstruction than the one already crossed; when $\rho_+ \geq r_c$ there is no obstruction on the z_+ side at all, which is only more favorable. The theorem does *not* identify the exact transition exponent between the two basins, which would require ruling out returns to the initial basin and competing collective paths in $\mathbb{R}^n$.

Remark 3.6 (Comparison with independent Langevin dynamics). For a single overdamped Langevin particle in the potential f started at z_-, the classical small-noise exit exponent from the basin of z_- is the physical barrier $h = f(s) - f(z_-)$. For n *independent* copies, the first of the n exit times has the same exponential scale h: taking a minimum over n i.i.d. variables with common exponential rate h/σ changes only the polynomial-in-n prefactor, not the exponent, as $\sigma \downarrow 0$.

By (16), the channel construction gives $V_\beta(\Gamma(z_-), \Gamma(s)) \leq u_h < h$ for the joint Softmin diffusion, in *every* regime of h (see Remark 3.4). This is a comparison of an *upper bound on a quasipotential* for the interacting system against the *exact* exponent h of the noninteracting one; it shows the admissible channel is exponentially cheaper than independent escape, not that the true joint exit exponent is smaller. The stronger, exit-*time* statement that the interaction cannot slow escape holds only under the hypotheses of Corollary 3.3, which are not verified in this paper.

Remark 3.7 (Higher dimensions and annealing). The channel identity of Part 1 extends algebraically to $d > 1$: at a critical point s of f the active Hessian block is $u'(h)\nabla^2 f(s)$, so if s is an index-one saddle and $u'(h) < 0$ the induced local minimum is a genuinely one-dimensional phenomenon, and transverse unstable directions may open cheaper escape routes. Regarding temperature, the fixed-β bound (15) decreases like β^{-1}, favoring large β for barrier reduction; but large β also suppresses the drift of non-leading particles through their Gibbs weights. A decaying schedule β_t is therefore natural, and is used heuristically in the experiments; its rigorous large-deviation analysis is left open.

4 Numerical Experiments

We evaluate the capacity of Softmin dynamics to escape local minima by comparing fixed ($\beta = 10$) and decaying schedules against Simulated Annealing (SA), Gradient-Aware Consensus-Based Optimization (CBO) [9], and Gradient-Enhanced Particle Swarm Optimization (PSO) [23]. The continuous dynamics are discretized using the Euler-Maruyama scheme. Landscape formulations, hyperparameters, initialization bounds, and hardware specifications are detailed in Appendix C; the source code is available at [3]. Because all evaluated methods require $O(n)$ gradient computations per step, they share an identical per-iteration computational cost. Consequently, the advantage of Softmin lies not in cheaper local updates, but in exponentially shorter transition times across topological barriers.

We first verify the large deviation scaling of Theorem 3.2 on a 1D double well ($\sigma = 0.05$). To isolate the exit phase, all particles are initialized in the left-most minimum. Results are shown in Appendix C.4.

To evaluate topological escape ($\sigma = 1$), subsequent benchmarks initialize the swarm entirely within a suboptimal basin to test each method's capacity to generate exploratory variance before reaching the maximum step limit. Figure 1 (Left) reports the 1D performance across varying barrier heights h. While Softmin variants demonstrate a stable, flat trajectory, CBO fails due to premature consensus, and SA exhibits severe exponential scaling. At extreme barriers ($h > 6.5$), SA and PSO display anomalous non-monotonicity due to a numerical discretization artifact: steep outer walls induce large discrete gradient steps, causing SA to artificially overshoot the barrier and PSO to escape via inertia. Softmin prevents this numerical instability through its intrinsically bounded pre-factor $A_{t,k}$.

Figure 1 (Center) extends this analysis to an unbalanced 2D quadruple well. In this higher-dimensional setting, the numerical artifacts observed in 1D vanish; particles subjected to large discrete gradient

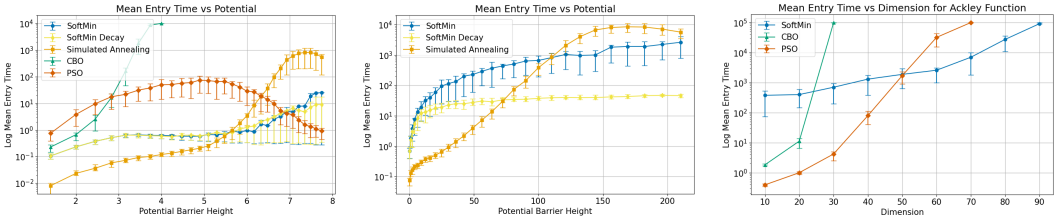


Figure 1: Numerical validation of the Softmin dynamics. Left: 1d double well benchmark. Center: 2d quadruple well benchmark (CBO and PSO omitted due to complete variance collapse). Right: Dimensional scalability on the generalized Ackley function, max steps allowed 10^5 (SA omitted due to maximum steps reached at $d = 10$). The error bars in the plots are drawn in correspondence to the quartiles of the distribution of the computed exit times.

updates typically scatter off steep walls rather than traversing the narrow topological saddle. Consequently, both PSO and CBO fail to escape the suboptimal well due to variance collapse and are omitted from the plot. Simulated Annealing again exhibits exponential scaling with barrier height, eventually plateauing when the required transition time exceeds the maximum step limit. In contrast, the Softmin methods outperform baseline techniques by actively repelling particles to generate the spatial variance necessary for escape. Furthermore, the advantage of the geometrically decaying schedule is evident: while fixed Softmin ($\beta = 10$) exhibits a slight upward trend at extreme barriers, the decaying variant maintains consistently lower transition times.

Finally, Figure 1 (Right) evaluates scalability on the generalized Ackley function across 10 to 90 dimensions. Particles are initialized uniformly over a compact domain containing an exponential number of local minima. SA fails to converge across all tested dimensions. As the search space volume increases, CBO and PSO reach the computational step limit without escaping by dimensions 30 and 70, respectively. In contrast, Softmin scales robustly. By actively repelling particles from suboptimal basins, the dynamics effectively drive the swarm toward the global optimum, demonstrating its efficacy in complex, high-dimensional landscapes.

5 Conclusion

In this work, we introduced a novel method for escaping local minima in optimization problems, based on the Softmin Energy interaction dynamics. We showed that the dynamics is able to escape local minima and find global optima by lowering effective potential barriers, leading to faster transition times between local minima in the small noise regime. We also proved that the convergence rate of the dynamics is faster than the one of a particle following the Langevin dynamics. Through numerical experiments, we validated our theoretical results and demonstrated the effectiveness of our method on benchmark functions such as double wells and the Ackley function. In the analyzed benchmarks, our method outperforms Simulated Annealing, Gradient-Enhanced Particle Swarm Optimization, and Gradient-Aware Consensus-Based Optimization in terms of transition times and convergence rates.

5.1 Limitations and Future Works

Our theoretical analysis provides rigorous guarantees in the practically relevant small-noise regime where the Friedlin-Wentzell Large Deviation Principles apply. While the exponential bound in Theorem 3.2 is not sharp, leaving room for a tighter characterization of the exponential rate in the regime where $\overline{V}_\beta$ scales quadratically in f^*, together with intermediate-noise asymptotics, the framework establishes fundamental advantages of particle cooperation for barrier reduction. Our equilibrium characterization (Theorem 2.1) and ensemble-descent estimate (Theorem 2.4) already require no convexity assumption on f; what remains open is sharper. The exit-time consequence of Theorem 3.2 holds unconditionally only at the level of the quasipotential, and becomes an actual statement about $\mathbb{E}\tau^\sigma$ only under the unverified metastable-domain hypotheses of Corollary 3.3; identifying the exact transition exponent between adjacent basins, and a rigorous large-deviation treatment of the annealed schedule β_t, remain open.

Future work should address several important directions. First, extending convergence proofs and explicit convergence rates to the non-convex regime, particularly for objectives with multiple disconnected basins, would significantly broaden applicability. Second, analyzing the mean-field limit as $n \to \infty$ could provide insights into the scaling of the method with ensemble size and potentially yield dimension-free convergence rates. Finally, tighter estimates for the exponential rate in the small noise regime can sharpen the theoretical bound to match experimental evidence more closely.

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A Freidlin-Wentzell Large Deviation Theory

This appendix summarizes the background large deviation theory used by Theorem 3.2 and Corollary 3.3. Consider a small-noise gradient diffusion on $\mathbb{R}^N$,

$$\mathrm{d}\mathbf{X}_t = -\nabla U(\mathbf{X}_t)\mathrm{d}t + \sqrt{2\sigma}\mathrm{d}B_t,$$

for a potential $U \in C^2(\mathbb{R}^N)$ and standard N-dimensional Brownian motion $(B_t)_{t\geq 0}$; in our application $N = nd$ and $U = nJ_\beta$. As $\sigma \downarrow 0$, sample paths of this diffusion concentrate around trajectories of the noiseless flow $\dot{\mathbf{X}}_t = -\nabla U(\mathbf{X}_t)$, and rare deviations from this flow are governed by the Freidlin-Wentzell large deviation principle [14]. For an absolutely continuous path $\phi : [0, T] \to \mathbb{R}^N$, the Freidlin-Wentzell action is

$$I_{0T}(\phi) := \frac{1}{4}\int_0^T \left\|\dot{\phi}(t) + \nabla U(\phi(t))\right\|^2 \mathrm{d}t, \tag{19}$$

and the quasipotential between a point $x \in \mathbb{R}^N$ and a closed set $A \subset \mathbb{R}^N$ is

$$V(x, A) := \inf_{T>0} \inf_{\substack{\phi(0)=x \\ \phi(T)\in A}} I_{0T}(\phi). \tag{20}$$

Intuitively, $I_{0T}(\phi)$ penalizes deviation from the noiseless flow, vanishing exactly along trajectories of $\dot{\phi} = -\nabla U(\phi)$; $V(x, A)$ is the cheapest such penalty over all paths from x reaching A, and quantifies how atypical (large-deviation-improbable) it is for the diffusion to travel from x to A without the help of noise.

Under the standard hypotheses of Freidlin-Wentzell exit theory [14] – in particular that D is a bounded domain with $x \in D$, that x is the unique asymptotically stable equilibrium of the noiseless flow in D, and that ∂D contains no other equilibria – the exit time τ_D^σ of the diffusion from D, started at x, satisfies

$$\lim_{\sigma\downarrow 0} \sigma \log \mathbb{E}_x \tau_D^\sigma = V(x, \partial D). \tag{21}$$

For a gradient drift, the action of any path admits the exact decomposition

$$I_{0T}(\phi) = U(\phi(T)) - U(\phi(0)) + \frac{1}{4}\int_0^T \left\|\dot{\phi}(t) - \nabla U(\phi(t))\right\|^2 \mathrm{d}t, \tag{22}$$

obtained by expanding the square in (19) and completing it against $\mathrm{d}U(\phi(t))/\mathrm{d}t = \langle \nabla U(\phi(t)), \dot{\phi}(t)\rangle$. Consequently, any segment of ϕ that follows the noiseless flow *downhill* (i.e. $\dot{\phi} = -\nabla U(\phi)$) contributes zero action, while a segment moving *uphill* against the flow (i.e. $\dot{\phi} = \nabla U(\phi)$) costs exactly the potential increase along that segment. This identity is what allows the quasipotential of an explicit admissible path to be computed exactly, rather than merely bounded, in the proof of Theorem 3.2.

B Technical Proofs

Here we will collect some of the more technical proofs of the lemmas stated in the main body.

B.1 Proofs of Section 2

Lemma B.1 (Component gradient and Hessian at critical configurations)**.** *Let* $w_k = e^{-\beta f(x_k)}(\sum_j e^{-\beta f(x_j)})^{-1}$ *denote the Gibbs weight of particle* x_k*, and let* $A_{t,k} = nw_k\left[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))\right]$ *be the dynamic pre-factor. Then*

$$\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_{t,k}\nabla f(x_k), \qquad \sum_{k=1}^n A_{t,k} = n. \tag{23}$$

Moreover, for $k \neq \ell$ *the mixed Hessian block is the outer product*

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_\ell \partial x_k} = \nabla f(x_k)\big(\nabla_{x_\ell} A_{t,k}\big)^\top, \tag{24}$$

which vanishes whenever $\nabla f(x_k) = 0$. Consequently, if $\mathbf{x}$ is such that $\nabla f(x_k) = 0$ for every k, then $\nabla(nJ_\beta(\mathbf{x})) = 0$ and the Hessian is block diagonal,

$$\nabla^2(nJ_\beta(\mathbf{x})) = \operatorname{blockdiag}\big(A_{t,1}\nabla^2 f(x_1), \dots, A_{t,n}\nabla^2 f(x_n)\big). \tag{25}$$

Finally, if $A_{t,k} = 0$ then $\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = -n\beta w_k \nabla f(x_k)\nabla f(x_k)^\top$.

Proof. We compute the derivatives using the chain rule and the properties of the Gibbs weights. First, we establish the gradient of the weight w_i with respect to x_k:

$$\nabla_{x_k} w_i = \begin{cases} \beta w_i w_k \nabla f(x_k) & \text{if } i \neq k, \\ -\beta w_k(1 - w_k)\nabla f(x_k) & \text{if } i = k. \end{cases}$$

For the first derivative of $nJ_\beta(\mathbf{x}) = n\sum_i w_i f(x_i)$, an identical computation to the main text gives $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = nw_k[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]\nabla f(x_k) = A_{t,k}\nabla f(x_k)$; summing over k and using $\sum_k w_k = 1$, $\sum_k w_k f(x_k) = J_\beta(\mathbf{x})$, gives $\sum_k A_{t,k} = n[1 - \beta(J_\beta(\mathbf{x}) - J_\beta(\mathbf{x}))] = n$.

For the mixed second derivative ($k \neq \ell$), we differentiate $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_{t,k}\nabla f(x_k)$ with respect to x_ℓ. Because $\nabla f(x_k)$ is independent of x_ℓ, the Jacobian matrix is the outer product $\nabla f(x_k)(\nabla_{x_\ell} A_{t,k})^\top$. Substituting $\nabla_{x_\ell} w_k = \beta w_k w_\ell \nabla f(x_\ell)$ and $\nabla_{x_\ell} J_\beta(\mathbf{x}) = \frac{1}{n} A_{t,\ell}\nabla f(x_\ell)$,

$$\begin{aligned} \nabla_{x_\ell} A_{t,k} &= n(\nabla_{x_\ell} w_k)\left[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))\right] + nw_k\beta\left(\nabla_{x_\ell} J_\beta(\mathbf{x})\right) \\ &= \beta w_\ell\big(nw_k[1 - \beta(f(x_k) - J_\beta(\mathbf{x}))]\big)\nabla f(x_\ell) + \beta w_k A_{t,\ell}\nabla f(x_\ell) \\ &= \beta\left[w_\ell A_{t,k} + w_k A_{t,\ell}\right]\nabla f(x_\ell), \end{aligned}$$

which gives (24) and vanishes whenever $\nabla f(x_k) = 0$ (or symmetrically $\nabla f(x_\ell) = 0$). If $\nabla f(x_k) = 0$ for every k, every off-diagonal block vanishes and (23) gives $\nabla(nJ_\beta(\mathbf{x})) = 0$; the diagonal block computed below then reduces to $A_{t,k}\nabla^2 f(x_k)$, giving the block-diagonal form (25).

Finally, for the diagonal block ($k = \ell$), applying the product rule to $\nabla_{x_k}(nJ_\beta(\mathbf{x})) = A_{t,k}\nabla f(x_k)$ includes the local Hessian $\nabla^2 f(x_k)$:

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = A_{t,k}\nabla^2 f(x_k) + \nabla f(x_k)\big(\nabla_{x_k} A_{t,k}\big)^\top.$$

Using $\nabla_{x_k} w_k = -\beta w_k(1 - w_k)\nabla f(x_k)$ and $\nabla_{x_k} J_\beta(\mathbf{x}) = \frac{1}{n} A_{t,k}\nabla f(x_k)$, a computation identical in structure to the mixed case gives

$$\nabla_{x_k} A_{t,k} = -\beta\left[nw_k + A_{t,k} - 2w_k A_{t,k}\right]\nabla f(x_k),$$

so that, substituting back,

$$\frac{\partial^2(nJ_\beta(\mathbf{x}))}{\partial x_k^2} = A_{t,k}\nabla^2 f(x_k) - \beta\left[nw_k + A_{t,k} - 2w_k A_{t,k}\right]\nabla f(x_k)\nabla f(x_k)^\top,$$

which reduces to $-n\beta w_k \nabla f(x_k)\nabla f(x_k)^\top$ when $A_{t,k} = 0$. □

Proof of Theorem 2.1. A locally asymptotically stable equilibrium of the gradient flow (3) is an isolated local minimum of nJ_β: indeed $\frac{d}{dt}(nJ_\beta(\mathbf{x}(t))) = -\|\nabla(nJ_\beta)\|^2 \leq 0$, so if points arbitrarily close to $\bar{\mathbf{x}}$ had strictly smaller value the trajectories starting there could not converge to $\bar{\mathbf{x}}$; and a non-isolated equilibrium admits nearby constant trajectories that likewise do not converge to $\bar{\mathbf{x}}$.

Fix k and freeze the other particles at $\bar{x}_j$, $j \neq k$. Since nJ_β depends on x_k only through $r = f(x_k)$, we may write

$$nJ_\beta(\bar{x}_1, \dots, x_k, \dots, \bar{x}_n) = g(f(x_k)), \qquad g(r) := n\,\frac{re^{-\beta r} + C_1}{e^{-\beta r} + C_0},$$

with $C_0 := \sum_{j\neq k} e^{-\beta f(\bar{x}_j)} > 0$ and $C_1 := \sum_{j\neq k} f(\bar{x}_j)e^{-\beta f(\bar{x}_j)}$. By the chain rule and Lemma B.1, wherever $\nabla f(x_k) \neq 0$ one has $g'(f(x_k)) = A_{t,k}$; by continuity of both sides in $f(x_k)$, $g'(f(\bar{x}_k)) = A_{t,k}$ evaluated at $\bar{\mathbf{x}}$. Writing $g'(r) = n\omega(r)[1 - \beta(r - g(r)/n)]$ with $\omega(r) = e^{-\beta r}/(e^{-\beta r} + C_0) \in$

$(0,1)$, a direct computation shows that at any critical point r_0 of g (where the bracket vanishes), $g''(r_0) = -n\beta\omega(r_0) < 0$.

Step 1: $A_{t,k} \neq 0$. Suppose $A_{t,k} = 0$ at $\bar{\mathbf{x}}$. By the above, $r_0 = f(\bar{x}_k)$ is a critical point of g, and it is a strict local maximum. If f is non-constant near $\bar{x}_k$, there are points y arbitrarily close to $\bar{x}_k$ with $f(y)$ close to but different from r_0, whence $nJ_\beta(\dots, y, \dots) = g(f(y)) < g(r_0) = nJ_\beta(\bar{\mathbf{x}})$, contradicting local minimality. If f is constant near $\bar{x}_k$, moving x_k within that level set leaves every objective value, Gibbs weight, and gradient unchanged, producing a continuum of equilibria arbitrarily close to $\bar{\mathbf{x}}$ and contradicting isolation. Hence $A_{t,k} \neq 0$.

Step 2: $\nabla f(\bar{x}_k) = 0$. Since $A_{t,k} \neq 0$ at $\bar{\mathbf{x}}$, $g'(f(\bar{x}_k)) \neq 0$, so g is strictly monotone near $f(\bar{x}_k)$. If $\nabla f(\bar{x}_k) \neq 0$, then f takes values on both sides of $f(\bar{x}_k)$ arbitrarily near $\bar{x}_k$; strict monotonicity of g then yields nearby configurations with $nJ_\beta < nJ_\beta(\bar{\mathbf{x}})$, contradicting local minimality. Hence $\nabla f(\bar{x}_k) = 0$.

Step 3: classification. If $A_{t,k} > 0$ at $\bar{\mathbf{x}}$, then g is strictly increasing near $f(\bar{x}_k)$, so local minimality of nJ_β forces $f(x_k) \geq f(\bar{x}_k)$ for x_k near $\bar{x}_k$; thus $\bar{x}_k$ is a local minimum of f. If $A_{t,k} < 0$, the symmetric argument shows $\bar{x}_k$ is a local maximum. Because $nw_k > 0$, the sign of $A_{t,k}$ is that of $1 - \beta(f(\bar{x}_k) - J_\beta(\bar{\mathbf{x}}))$, giving the stated threshold. Finally, $\sum_k A_{t,k} = n$ forces some $A_{t,k} > 0$ at $\bar{\mathbf{x}}$, so at least one particle is at a local minimum of f. □

Proof of Theorem 2.4. Fix t and let i attain the minimum, $f(x_i(t)) = m(t)$. Since $f(x_i(t)) \leq f(x_j(t))$ for all j, we have $e^{-\beta_t f(x_j(t))} \leq e^{-\beta_t f(x_i(t))}$, hence $\sum_j e^{-\beta_t f(x_j(t))} \leq n e^{-\beta_t f(x_i(t))}$ and therefore $nw_i(t) \geq 1$. As $J_{\beta_t}(\mathbf{x}(t))$ is a convex combination of the values $f(x_j(t))$, it satisfies $J_{\beta_t}(\mathbf{x}(t)) \geq f(x_i(t))$, so $1 - \beta_t(f(x_i(t)) - J_{\beta_t}(\mathbf{x}(t))) \geq 1$. Multiplying,

$$A_{t,i} \geq 1 \qquad \text{for every } i \text{ attaining the minimum at } t. \tag{26}$$

Along such a particle,

$$\frac{d}{dt} f(x_i(t)) = \langle \nabla f(x_i(t)), \dot{x}_i(t) \rangle = -A_{t,i} \|\nabla f(x_i(t))\|^2 \leq -\|\nabla f(x_i(t))\|^2.$$

The map $m = \min_k g_k$, with $g_k(t) = f(x_k(t)) \in C^1$, is locally Lipschitz, hence absolutely continuous and differentiable for a.e. t. At a differentiability point t, choose i attaining the minimum; the differentiable function $g_i - m \geq 0$ vanishes at t and thus has zero derivative there, giving $m'(t) = g_i'(t)$. Combined with the previous display,

$$m'(t) = -A_{t,i} \|\nabla f(x_i(t))\|^2 \leq -\|\nabla f(x_i(t))\|^2 \leq 0 \qquad \text{for a.e. } t.$$

Integrating, m is nonincreasing, so every minimizing particle stays in $S_0 = \{x : f(x) \leq m(0)\}$, where the Polyak–Łojasiewicz condition (Definition 2.3) applies:

$$m'(t) \leq -\|\nabla f(x_i(t))\|^2 \leq -2\mu\big(f(x_i(t)) - f_\star\big) = -2\mu\big(m(t) - f_\star\big).$$

Setting $y(t) := e^{2\mu t}(m(t) - f_\star)$, which is absolutely continuous, we get $y'(t) = e^{2\mu t}[m'(t) + 2\mu(m(t) - f_\star)] \leq 0$ for a.e. t, so $y(t) \leq y(0)$, i.e.

$$m(t) - f_\star \leq e^{-2\mu t}\big(m(0) - f_\star\big), \qquad t \geq 0. \qquad \square$$

B.2 Proofs of Section 3

Proof of Lemma 3.1. Write $r := r(x)$. Factoring $e^{-\beta f(z_-)}$ out of the Gibbs weights at $\Gamma(x)$,

$$w_1(\Gamma(x)) = \frac{e^{-\beta r}}{n - 1 + e^{-\beta r}}, \qquad w_j(\Gamma(x)) = \frac{1}{n - 1 + e^{-\beta r}} \quad (j \geq 2),$$

whence $J_\beta(\Gamma(x)) - f(z_-) = re^{-\beta r}/(n - 1 + e^{-\beta r}) = u(r)/n$, which is (11). Therefore

$$f(x) - J_\beta(\Gamma(x)) = r - \frac{re^{-\beta r}}{n - 1 + e^{-\beta r}} = \frac{(n-1)r}{n - 1 + e^{-\beta r}},$$

and substituting into $A_{t,1} = nw_1[1 - \beta(f(x) - J_\beta)]$ gives

$$A_{t,1}(\Gamma(x)) = \frac{ne^{-\beta r}}{n - 1 + e^{-\beta r}} \cdot \frac{n - 1 + e^{-\beta r} - \beta(n-1)r}{n - 1 + e^{-\beta r}} = \frac{ne^{-\beta r} q(r)}{(n - 1 + e^{-\beta r})^2}.$$

Differentiating (11) directly yields the same expression for $u'(r)$, proving (12). In particular $A_{t,1}(\Gamma(x)) = 0$ if and only if $q(r) = 0$, which by the displayed formula for $f(x) - J_\beta$ is equivalent to $f(x) - J_\beta(\Gamma(x)) = 1/\beta$, i.e. to the vanishing of the Softmin threshold gap of the active particle.

Since $q'(r) = -\beta(n-1) - \beta e^{-\beta r} < 0$ for all $r \in \mathbb{R}$, $q(0) = n > 0$, and $q(r) \to -\infty$ as $r \to \infty$, q has a unique zero $r_c > 0$. From $q(1/\beta) = e^{-1} > 0$ we get $r_c > 1/\beta$; and $q(r_c) = 0$ reads $(n-1)(\beta r_c - 1) = e^{-\beta r_c} < 1$, giving $\beta r_c < n/(n-1)$. This is (13). As $ne^{-\beta r}/(n-1+e^{-\beta r})^2 > 0$, the sign of u' is the sign of q, so u is strictly increasing on $(-\infty, r_c]$ and strictly decreasing on $[r_c, \infty)$. □

Proof of Theorem 3.2. Part 1. For $r \le 0$ we have $u(r) \le 0$. For $r > 0$, since $n - 1 + e^{-\beta r} > n - 1$,

$$u(r) < \frac{n}{n-1} r e^{-\beta r} \le \frac{n}{n-1} \max_{\rho \ge 0} \rho e^{-\beta \rho} = \frac{n}{e(n-1)\beta},$$

the maximum being attained at $\rho = 1/\beta$. Hence $u_c = \max_{\mathbb{R}} u < n/(e(n-1)\beta)$, which is (15); the bound on r_c is (13).

Part 2. A channel path has the form $\phi(t) = \Gamma(\psi(t))$; the frozen coordinates stay at the critical point z_-, so their drift vanishes and, with $W(x) := u(r(x))$, the action reduces to $I_{0T} = \frac{1}{4}\int_0^T |\dot\psi + W'(\psi)|^2 \mathrm{d}t$. The one-dimensional version of (22) is

$$I_{0T} = W(\psi(T)) - W(\psi(0)) + \frac{1}{4}\int_0^T |\dot\psi - W'(\psi)|^2 \mathrm{d}t.$$

By (8), r is a strictly increasing bijection from $[z_-, s]$ onto $[0, h]$, so there is a unique $x_m \in (z_-, s]$ with $r(x_m) = \min\{h, r_c\}$. On $[z_-, x_m]$ we have $r(x) \le r_c$, so by Lemma 3.1 W is strictly increasing there, with total rise $W(x_m) - W(z_-) = u_h$; on $[x_m, s]$ we have $r(x) \ge r_c$, so W is nonincreasing. Every continuous channel path from z_- to s passes through x_m, and applying the displayed decomposition to the sub-path up to the first hitting time of x_m bounds the action from below by u_h. This value is attained in the quasipotential sense by following the reversed deterministic flow from z_- up to x_m (cost exactly u_h) and then the forward flow toward s at zero additional cost, with the standard finite-time approximations near the critical points. Therefore the channel quasipotential equals u_h, and since V_β is an infimum over the larger class of all paths in $\mathbb{R}^n$,

$$V_\beta(\Gamma(z_-), \Gamma(s)) \le u_h.$$

Because $u(r)/r = ne^{-\beta r}/(n - 1 + e^{-\beta r}) < 1$ for $r > 0$, we have $u_h < \min\{h, r_c\} \le h$, which gives (16).

Part 3. Assume $r_c < h$. At $\Gamma(s)$ every particle sits at a critical point of f (either s or z_-). For the active particle, (12) gives $A_{t,1}(\Gamma(s)) = u'(h) < 0$ because $h > r_c$, and $f''(s) < 0$ since s is a nondegenerate maximum. For a frozen particle at z_-,

$$A_{t,j}(\Gamma(s)) = n w_j(\Gamma(s))\big[1 + \beta\big(J_\beta(\Gamma(s)) - f(z_-)\big)\big] > 0$$

by (11), since $J_\beta(\Gamma(s)) - f(z_-) = u(h)/n > 0$, and $f''(z_-) > 0$. Hence Corollary 2.2 applies and $\Gamma(s)$ is a strict local minimum of nJ_β. On the z_- side, r decreases strictly from h through r_c as x moves from s toward z_- (recall $x_m \in (z_-, s)$ is where $r(x_m) = r_c$, well-defined since $r_c < h$), so the channel well at $\Gamma(s)$ has barrier $u_c - u(h)$ on this side unconditionally — this uses only $r_c < h$, not ρ_+.

The z_+ side depends on ρ_+. On (s, z_+), r decreases strictly from h to ρ_+ by (8)–(9).

If $\rho_+ < r_c$, then r crosses r_c on this interval, so there is a unique ξ_+ with $r(\xi_+) = r_c$. Between s and ξ_+ the reduced potential rises from $u(h)$ to u_c; the gradient-action argument of Part 2 gives channel action $u_c - u(h)$, and since $0 < u(h) < u_c$ we obtain (18), so the well is symmetric. Finally, for $x \in (\xi_+, z_+)$ we have $r(x) < r_c$, hence $A_{t,1}(\Gamma(x)) = u'(r(x)) > 0$ by (12), while $f'(x) < 0$; therefore the active coordinate of the deterministic channel flow satisfies $\dot x = -A_{t,1}(\Gamma(x)) f'(x) > 0$, driving the particle toward z_+.

If instead $\rho_+ \ge r_c$, then $r(x) \in [r_c, h)$ for every $x \in (s, z_+)$: the interval is entirely on the decreasing side of u, where u and r are both strictly decreasing in x, so $u(r(x))$ is strictly increasing in x throughout (s, z_+), from $u(h)$ to $u(\rho_+)$. No interior peak occurs, so no point ξ_+ with $r(\xi_+) = r_c$ exists in (s, z_+), and the channel presents no barrier separating $\Gamma(s)$ from $\Gamma(z_+)$ on this side.

If instead $h \leq r_c$, then $r(x) \leq h \leq r_c$ for x near s, with equality only at $x = s$, and the strict monotonicity of u on $(-\infty, r_c]$ shows that $x \mapsto u(r(x))$ has a strict local maximum at s. Hence $\Gamma(s)$ is not a local minimum of nJ_β. □

C Appendix: Experimental Details and Hyperparameters

To ensure full reproducibility of the numerical experiments presented in Section 4, this appendix details the hyperparameters and initialization schemes used across all benchmark landscapes. All continuous-time dynamics were discretized using the Euler-Maruyama method. The source code used to execute these experiments is publicly available at [3].

C.1 Hardware and Computational Resources

All numerical experiments were executed on the *Toeplitz* Scientific Computing Cluster hosted by the Department of Mathematics at the University of Pisa. To fully leverage the vectorized nature of the simulations and evaluate independent trials in parallel, jobs were strictly allocated to a single compute node at a time. The cluster nodes utilized for these experiments include:

- **High-Capacity Nodes:** AMD EPYC 7763 processors featuring 128 physical cores (64 cores per socket, 2 sockets, 256 logical threads) and 2 TB of RAM.
- **Standard Nodes:** Intel Xeon CPU E5-2650 v4 @ 2.20GHz featuring 24 physical cores (12 cores per socket, 2 sockets) and 256 GB of RAM. These nodes were specifically utilized to parallelize the independent runs of the experiments using Python's `multiprocessing` framework.

C.2 General Experimental Settings

Table 1 summarizes the shared parameters utilized across all algorithms for each specific landscape. The swarm size and maximum step limits were kept constant to provide a fair baseline for evaluating the topological escape mechanisms.

Table 1: General settings and landscape configurations.

Parameter	**Value / Description**
Swarm Size (n)	100
Independent Runs	100
Max Step Limit	10,000,000
1D Double Well	$f(x) = \frac{1}{4}x^4 - ax^2$
Initialization	Uniformly sampled around the left-most minimum
Domain Bounds	$[-\infty, \infty]$
2D Quadruple Well	$f(x, y) = x^4 + y^4 - a(x^2 + y^2) - xy$
Initialization	Uniformly sampled around the suboptimal trap
Domain Bounds	$[-4.0, 4.0]^2$
High-Dimensional Ackley	Generalized Ackley ($d \in [10, 90]$)
Function Parameters	$a = 20.0$, $b = 0.2$, $c = 2\pi$
Initialization	Uniformly sampled in $[-10, 10]^d$
Domain Bounds	$[-10, 10]^d$

C.3 Algorithm-Specific Hyperparameters

The following tables detail the algorithm-specific hyperparameters. For methods utilizing a decaying schedule (Softmin Decaying, Simulated Annealing), the relevant parameter was updated at each step t according to a geometric decay schedule of the form $P_t = P_0 \times (\text{decay rate})^t$.

Table 2: Hyperparameters for Softmin Energy Dynamics.

Parameter	**Value**
Learning Rate (η)	0.01
Adaptive Learning Rate (Adagrad)	Disabled (False)
Noise Intensity (σ)	1.0 (Benchmarks), 0.05 (Theorem 3.2 Verification)
Noise Distribution	Normal
Fixed Inverse Temperature (β)	10.0
Decaying Inverse Temperature (β_0)	10.0
Geometric Decay Rate for β	0.9999

Table 3: Hyperparameters for Simulated Annealing (SA).

Parameter	**Value**
Learning Rate (η)	0.01
Base Noise Variance Factor	$2 \times \sigma \times \eta$
Initial Temperature (T_0)	10.0
Cooling Schedule	Geometric
Geometric Cooling Rate	0.9999

Table 4: Hyperparameters for Gradient-Aware Particle Swarm Optimization (PSO).

Parameter	**Value**
Gradient Step Learning Rate (η)	0.01
Inertia Weight (w)	0.729
Cognitive Coefficient (c_1)	1.49445
Social Coefficient (c_2)	1.49445
Injected Noise Floor	0.5

Table 5: Hyperparameters for Consensus-Based Optimization (CBO).

Parameter	**Value**
Learning Rate (Δt)	0.01
Consensus Weight (λ)	1.0
Exploration Noise (σ_{CBO})	$\sqrt{2}$
Temperature (α)	10.0

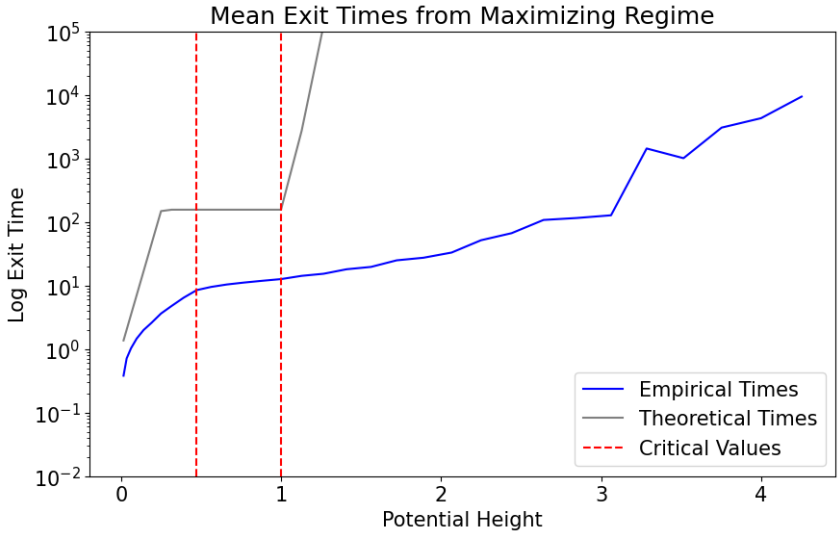


Figure 2: Empirical exit times strictly follow the bound from Theorem 3.2

C.4 Further Experimental Results

Figure 2 compares the empirical exit times against the theoretical upper bound $\overline{V}_\beta$. Dashed lines demarcate phase transitions where the dominant term governing the exit time shifts: from f^* to $n/(2(n-1)\beta)$, and ultimately to $(\beta/2)(f^* - 1/\beta)^2$. Validating our theoretical predictions, the empirical slope changes exactly at these boundaries, most prominently at the first transition. Furthermore, while the empirical times remain bounded by $\overline{V}_\beta$ across all regimes, the theoretical bound is not tight in the third phase.